

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 189

THE SPACE MICROWAVE THE CLAMPED-PLATE ROTATOR

nothing need change in design much
the spring seats what gravity did



hot food is morale — morale is mission hardware

4-PRONG HOLDERS · SPRING CLAMP · KEPT ROTATION

CONTAINED, ROTATED, EVEN-HEATED

Crew life — v1.0 in force

GP-IM-189

Revision v1.0 — 23 August 2026

190 of 290

VETTED BATCH 23 — PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 189

PHASE 189: THE SPACE MICROWAVE — THE CLAMPED-PLATE ROTATOR (ZERO-G HOT FOOD DOCTRINE) — STATUS: CURRENT — PASS (VET BATCH 23).

The galley counterpart to Phase 177: the birthday cake table keeps dinner on the plate while you eat it; this oven keeps dinner on the plate while you cook it. His opening line is the design philosophy: nothing need change in design much — a standard microwave, minimally modified. The turntable is replaced by two 4-pronged plate holders, one top one bottom, one drives and one rotates; prong spread 100mm, slightly flexible, red silicone coated. The upper set rides a rod through a seal with a metal spring clamp — positive mechanical capture where gravity used to seat the plate. The plate assembly loads at the prep bench: the hole plate, the food, a plastic/silicone mesh circle, then the thin top-and-bottom cover set like hospital food, side-locked, holes in the top cover against expansion. The audit's rod-seal flag was answered same sitting: the turntable shaft penetration with its seal already exists in every production microwave — 'so they wont panic'. The vet passed it clean.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-189, v1.0 — 23 August 2026. The one hundred and ninetieth manual of the program — 190 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the oven law as seated (contents line, the bodies — the problem, the rotator, the plate assembly — the audit and provenance, verbatim) — §2 why clamp and rotate (graded) — §3 the system in the cavity — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 189: **The Space Microwave — The Clamped-Plate Rotator (Zero-G Hot Food Doctrine)**. A standard microwave minimally modified: turntable replaced by two 4-pronged plate holders (100mm spread, flexible red silicone coated) — lower set drives, upper set rotates, and the

upper set rides a spring-lift rod (1-inch lift, side-lever operated) clamping the plate top-and-bottom so rotation survives the loss of gravity. Plate assembly: bench recess for loading, silicone mesh circle on the food, thin hospital-tray clamshell with side lock and vent holes against expansion pressure. The drill: lever up, plate in, release, close, start.*'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Retention of Microwave Cooking in Zero-G — the plate is clamped, the food is covered, the rotation is kept [*WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%'*]

Crew Life Baseline: Zero-G Galley Hardware — Contained, Rotated, Even-Heated Meals

Live instruction, 30 July 2026. Third build of the sitting, and the galley counterpart to Phase 177: the birthday cake table keeps dinner on the plate while you eat it; this oven keeps dinner on the plate while you cook it. Archer's opening line is the design philosophy: nothing need change in design much — a standard microwave, minimally modified, instead of a bespoke space appliance.

THE PROBLEM — GRAVITY WAS THE LID (VERBATIM)

- **The Earth assumption:** a microwave turntable holds food by weight alone. Remove gravity and the food floats off the plate, the plate drifts off the turntable, liquids glob and roam the cavity, and the meal migrates to the walls mid-cycle — uneven heating, lost food, and a cleaning problem in a sealed hull.
- **The hidden second problem:** rotation is not optional. Microwave cavities heat in standing-wave patterns; turning the food through the pattern is what evens the cooking. A zero-G oven that merely restrains the food but cannot rotate it cooks hot and cold stripes. The design must clamp AND turn.

THE CLAMPED-PLATE ROTATOR (VERBATIM)

- **No glass plate:** the turntable is gone. In its place, two 4-pronged plate holders — one top, one bottom. The lower set drives; the upper set rotates with it. The prong spread is 100mm diameter, and the prongs are slightly flexible, red silicone coated — food-safe, heat-tolerant, and gripping without marring the plate.
- **The spring-lift upper set:** the upper prong set rides on a rod through a seal, with a metal spring inside the upper stand's metal casing giving a 1-inch lift. Press the small side lever to raise the upper set, seat the plate on the lower prongs, release the lever — the spring clamps the plate between the two prong sets. Positive capture, top and bottom, no gravity required.
- **Rotation preserved:** clamped between driven and idler prong sets, the plate turns through the cavity's heating pattern exactly as an Earth turntable meal does — the even-heating mechanism survives the loss of gravity intact.

THE PLATE ASSEMBLY & THE PREP BENCH (VERBATIM)

- **The bench recess:** on the food preparation bench, the plate sits in a hole in the bench while being loaded — the recess holds the plate in zero-G so prep is done with both hands and no drifting crockery.
- **The mesh circle:** a plastic/silicone mesh circle is placed on top of the loaded food — the first restraint, keeping portions seated during transfer from bench to oven.
- **The clamshell cover set:** a thin top-and-bottom cover set closes over the plate — the way hospital food arrives — the top virtually pressing on the mesh, locked by a plastic side lock. The top cover carries holes: they prevent expansion pressure building inside and let some heat and steam out. A vented cover, not a sealed one — by design.
- **The drill, as specified:** food to microwave; open door; press the small lever on the side to pull the spring up manually; place the plate in the centre on the lower prong set; release the lever; close the door; start. One hand holds the lever, one hand seats the plate — a two-handed drill, no loose steps, nothing left floating.

PHYSICS NOTE — THE AUDIT (KIMI: AFFIRMED — THE CLAMP SOLVES FLOAT, THE ROTATION SOLVES HOTSPOTS, THE VENTS SOLVE PRESSURE) (VERBATIM)

Every element answers a real term in the physics. The clamp answers zero-G: weight no longer seats the plate, so a spring does — positive mechanical capture, the same doctrine as Phase 177's pressure-differential dining, here in rigid form. The kept rotation answers the standing-wave problem: microwave cavities cook in fixed hotspot patterns, and moving the food through the pattern is the established even-heating mechanism — he preserved it instead of reinventing it, which is why 'nothing need change in design much' is credible. The vent holes answer steam physics: heating food generates vapour, and a sealed container in a microwave is a pressure vessel waiting to fail — vented clamshells relieve it, and the mesh beneath stops portions riding the vapour out. The hospital-tray lineage is genuine prior art, recorded: covered, compartmented, transported meal trays are a century-old institutional solution to moving food without spilling it. Flags, kept honest: (1) the upper rod passes through the cavity wall — any shaft penetration of a microwave cavity needs the standard choke/seal treatment against leakage; this is the same engineering that already passes turntable shafts through Earth ovens, flagged so it is built, not assumed. (2) Zero-G kills convection as well as gravity — heat distribution inside food and cavity differs from Earth, so cooking times are a bench-calibration item for the galley manual, not a number the file invents. (3) Prong silicone hardness and spring force are matched to plate mass and cover thickness — workshop dial-ins. None of the flags leave the crew hungry.

[Audit flag (1) answered by Archer, same sitting — 30 July 2026]: "actually, the rod seal already exists in the lower turning component in all microwaves so they wont panic" — recorded verbatim and affirmed: the

turntable shaft penetration with its seal is already in every production microwave, so the upper rod duplicates hardware the manufacturer already buys and certifies. The flag stands as the record of the question; this is the answer — no new component class, no certification panic.

ORIGIN OF THOUGHT — PHASE 189 (VERBATIM)

Archer, 30 July 2026: "space microwave oven. nothing need change in design much, the inner has no plate it has two 4 pronged plate holders , one top one bottom, one drives the other rotates, the prong spread is 100mm diameter, the are slightly flexible red silicone coated. the upper set is on a rod through a seal and a metal spring inside the upper stand metal casing it has a 1 inch lift."

"the food bench microchip bench where the food is carefully prepared has the hole plate there being loaded with food, a plastic/silicone mesh circle placed on top, then into a thin top and bottom cover set like hospital food arrives the top virtual pressing on the mesh, plastic side lock, the top cover has holes to prevent expanses and let out some heat. food to microwave, open door press small lever on side to pull up spring manually, place plate in centre on lower prong set, release lever, close door, start."

Why it matters, in plain English: hot food is morale, and morale is mission hardware — ask any crew six months from Earth. But the deeper entry is the philosophy in his first sentence: 'nothing need change in design much.' The fleet will not carry bespoke space appliances invented from a clean sheet; it will carry proven Earth machines with the gravity assumption engineered out of them — a spring where weight used to be, a clamp where a shelf used to be. That is the same doctrine as the wiper motor on the gasifier and the shoe lights on the battlestar: proven parts, re-assigned. The galley now closes its loop with the dining room: Phase 189 cooks it contained, Phase 177 serves it contained, and the steak-eating taxans eat hot.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CLAMP ANSWERS ZERO-G, AFFIRMED: weight no longer seats the plate, so a spring does — positive mechanical capture, the same doctrine as Phase 177's pressure-differential dining, here in rigid form. The prongs' slight flex and silicone coat grip the plate rim without point loads.

THE KEPT ROTATION ANSWERS THE STANDING-WAVE PROBLEM: microwave cavities cook in fixed hot and cold spots; gravity ovens solved it with a turntable and the solution survives the loss of gravity untouched. Rotation carries every part of the meal through the nodes — even heating is a geometry problem, not a gravity problem.

THE SEAL IS HERITAGE, AFFIRMED ON HIS ANSWER: the turntable shaft penetration and its seal exist in every production microwave; the upper rod duplicates a proven penetration instead of inventing one. 'Nothing need change in design much' is the fleet's appliance philosophy: proven Earth hardware, minimum modification, zero-G adaptations only where physics demands.

§3 — THE SYSTEM IN THE CAVITY

- **The plate holders:** two 4-pronged sets, top and bottom — one drives, one rotates; 100mm prong spread; slightly flexible, red silicone coated.
- **The clamp:** the upper set on a rod through a seal, metal spring — positive capture, the spring seats what gravity used to.
- **The plate assembly:** the hole plate loaded with food at the prep bench; the plastic/silicone mesh circle on top; the thin top-and-bottom cover set, hospital-food style, the top virtually pressing the mesh; plastic side lock; holes in the top cover against expansion.
- **The prep bench:** the food microchip bench where the plate is loaded — preparation stays bench-side, the oven stays sealed.
- **The doctrine line:** hot food is morale, and morale is mission hardware.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The galley triangle closes here: Phase 189 cooks it contained, Phase 177 serves it contained, Phase 190 washes it contained — containment is the zero-G galley law.
- The minimum-modification philosophy pairs with Phase 199's robust-over-new pull: proven hardware, adapted, beats bespoke hardware, invented.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 23 (16 August 2026 — phases 186-190, cross-checked in sitting 318), SEATED. The grade, verbatim from the batch: '189 → PASS.'

§6 — BUILD SEQUENCE

- 1. Strip a production microwave to the cavity; retain magnetron, door interlocks, and the turntable drive penetration with its seal.
- 2. Fit the two 4-pronged holders (100mm spread, silicone-coated flex prongs); wire one to the drive, let the other ride.
- 3. Fit the upper rod through the duplicated seal with the spring clamp.
- 4. Tool the plate assemblies: hole plates, mesh circles, cover sets with side locks and expansion holes.
- 5. Bench the cycle: clamp, rotate, cook; verify even heating and retention through ship-motion jolts.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 177 (the birthday cake table):** the serve-side half of the containment doctrine — vacuum-downforce dining where this oven is spring-clamped cooking.
- **Phase 190 (the dishwasher) and Phase 198 (the cake oven):** the wash and the bake complete the galley set; the plate assemblies are sized to the rack.

§8 — DO-NOT-BUILD

- Do NOT run an unclamped plate — the spring is the gravity substitute; no clamp, no cook.
- Do NOT seal the top cover without the expansion holes — hospital-tight is not the spec; the holes are law.
- Do NOT invent a bespoke oven — the doctrine is minimum modification of proven hardware; the seal panic was answered before it started.
- Do NOT prep at the oven — loading happens at the bench, covers on, before the plate sees the cavity.

§9 — OWED

OWED: the prong flex/silicone durometer sheet against plate-rim geometries; clamp retention under ship-motion jolt spectrum; even-heating survey across the rotation; the cover-set expansion-hole sizing. The audit's rod-seal flag stands answered verbatim in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-189, v1.0 — CURRENT — PASS (VET BATCH 23), seated law, master v2.1 (numerical print order). The one hundred and ninetieth manual of the program — 190 of 290. Built 23 August 2026, second of the 12-manual 'ok 12 more i will see what i get done in 30 mins to get to the 200' shift (the author's order, 23 August 2026, seated in sitting 597). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590 and 597): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the

200'. The phase's own chain, all seated in the master: the contents line, the masthead trio and the bodies (as seated); the live instruction of 30 July 2026 — third build of the sitting, the galley counterpart to Phase 177; the rod-seal flag answered same sitting ('so they wont panic'); the vet batch 23 clean pass in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the clamp retention and heating bench data, or his word.

END OF MANUAL — PHASE 189